AWS Certified AI Practitioner (AIF-C01) Full-Length Practice Exam

Instructor / Answer-Key Edition (Task 9 of 15)

This is original practice material created for this study program. It is not official AWS content and has not been produced, reviewed, or endorsed by Amazon Web Services. No question in this exam copies or closely mirrors any official AWS exam question.

Based on AIF-C01 Exam Guide Version 1.1 (April 30, 2026). Aligned to the official domain weightings: Domain 1 (20%), Domain 2 (24%), Domain 3 (28%), Domain 4 (14%), Domain 5 (14%).

PART 1: EXAM INSTRUCTIONS

Exam Name: AWS Certified AI Practitioner (AIF-C01) Practice Exam **Edition:** Instructor / Answer-Key Version (do not distribute this version to learners - see Task 10 for the clean learner version)

Exam Format

Detail	Value
Total questions	65
Recommended time	90 minutes
Question types	Multiple choice, multiple response, ordering, matching
Scoring on this practice exam	All 65 questions are scored (unlike the real exam, which scores 50 of 65 and treats 15 as unscored pilot questions)
Practice passing threshold	47/65 (73%)
Real exam passing threshold	700/1,000 scaled score (approximately 720/1,000 in practical terms)

How to Answer Each Question Type

- **Multiple choice:** Four answer options (A-D). Exactly one option is correct. Select the single best answer.
- **Multiple response:** Five or more answer options. The question will tell you how many correct answers to select ("Select TWO" or "Select THREE"). You must select all correct options and no incorrect options to receive credit. There is no partial credit.
- **Ordering:** You will be given a list of items and asked to place them in the correct sequence. All items must be placed in the correct order to receive credit.
- **Matching:** You will be given a list of concepts and a list of examples or definitions. Match each concept to its correct partner. All pairs must be correct to receive credit.

General Instructions

1. Set a 90-minute timer before you begin, to simulate real exam conditions.
2. Answer every question, even if uncertain. Unanswered questions are scored as incorrect, and there is no penalty for guessing.
3. Do not skip ahead to the answer key (Part 3) until you have completed all 65 questions.
4. After finishing, use the Domain Score Tracker (Part 5) to calculate your score by domain, and the Scoring Guide (Part 4) to interpret your overall readiness.
5. Questions are presented in mixed order across all five domains, exactly as they would appear on the real exam. They are not grouped by domain.

Reminder: This is original practice material designed to mirror the style, structure, and difficulty of the real AIF-C01 exam. It is a study tool, not official AWS content, and performance on this practice exam is not a guaranteed predictor of real exam performance.

PART 2: THE 65 PRACTICE QUESTIONS

Question 1 | Domain 2 | Difficulty: Intermediate

Place the following stages of the foundation model (FM) lifecycle in the correct order, from first to last.

1. Fine-tuning
2. Data selection
3. Deployment
4. Pre-training
5. Evaluation

Arrange items 1-5 into the correct sequence.

Question 2 | Domain 1 | Difficulty: Accessible

A retail operations manager describes a new system: “It uses layers of interconnected nodes to automatically learn patterns from thousands of historical images of damaged packages, without anyone manually defining what ‘dam-

A. Robotic process automation B. Deep learning C. Rules-based expert system D. Business intelligence dashboard

Question 3 | Domain 3 | Difficulty: Accessible

A startup needs a foundation model that supports both English and Spanish equally well, responds in under one second, and fits within a tight monthly budget. Which set of factors is the company evaluating?

A. Model training data labeling quality only
B. FM selection criteria, including multilingual support, latency, and cost
C. RLHF training stages
D. Diffusion model architecture

Question 4 | Domain 2 | Difficulty: Accessible

A developer explains that before a large language model can process a long document for a knowledge base, the document is split into smaller pieces, each piece is converted into a list of numbers that captures its meaning, and those numbers

A. Tokenization, then fine-tuning
B. Chunking, then embedding
C. Prompt engineering, then distillation
D. Context engineering, then pretraining

Question 5 | Domain 4 | Difficulty: Accessible

A company wants to ensure that its AI hiring tool produces truthful, accurate outputs and does not generate fabricated claims about a candidate's qualifications. Which core feature of responsible AI is most directly being addressed?

A. Veracity B. Inclusivity C. Robustness D. Confidence scoring

Question 6 | Domain 3 | Difficulty: Intermediate

A creative writing application wants its foundation model to generate more varied, surprising, and unconventional story ideas, even at the risk of occasional less coherent output. Which inference parameter adjustment

A. Decrease the temperature setting
B. Increase the temperature setting
C. Decrease the context window
D. Increase the number of fine-tuning epochs

Question 7 | Domain 5 | Difficulty: Accessible

A company runs a GenAI application using Amazon Bedrock. Under the AWS Shared Responsibility Model, which of the following is the customer's responsibility, not AWS's?

A. Securing the physical data centers that host Bedrock's infrastructure
B. Configuring IAM permissions and properly managing the data the customer sends to and stores within the application
C. Maintaining the underlying

Question 8 | Domain 1 | Difficulty: Accessible

A bank wants to score 2 million loan applications submitted over the past month and generate a single report tonight. There is no requirement for an immediate response to any individual application. Which type of inferencing

A. Real-time inferencing B. Batch inferencing C. Asynchronous inferencing D. Serverless inferencing

Question 9 | Domain 2 | Difficulty: Accessible

A team is building a chatbot using a foundation model and notices that when the conversation gets very long, the model starts to "forget" details mentioned earlier in the same conversation. What concept explains this behavior?

A. The model has reached its context window limit B. The model is experiencing hallucination C. The model has been fine-tuned incorrectly D. The model is using batch inferencing

Question 10 | Domain 3 | Difficulty: Accessible

A law firm wants its AI assistant to answer questions using the firm's own internal case files and precedent documents, rather than relying only on the model's general training data, so that answers are grounded in the firm's actual records. Which

A. Model distillation
B. Retrieval Augmented Generation (RAG)
C. Prompt poisoning
D. Reinforcement learning

Question 11 | Domain 4 | Difficulty: Accessible

A company wants to prevent its customer-facing GenAI chatbot from discussing competitor products or providing medical advice, regardless of how users phrase their requests. Which AWS capability is designed for this purpose?

A. Amazon SageMaker Clarify B. Amazon Bedrock Guardrails C. AWS CloudTrail D. Amazon Inspector

Question 12 | Domain 2 | Difficulty: Challenging

A marketing team is evaluating risks before rolling out a generative AI tool to draft customer-facing emails without human review. Select TWO disadvantages of GenAI that are most relevant to this decision.

A. Hallucination - the model may generate plausible-sounding but factually incorrect content B. GenAI models always require server infrastructure to be manually provisioned C. Non-determinism - the same prompt can produce differ-

Select TWO.

Question 13 | Domain 1 | Difficulty: Accessible

A hospital is building an AI system that will analyze MRI scans, doctors' handwritten notes, and a structured spreadsheet of patient lab values. Which statement correctly classifies these data types?

A. All three are structured data because they all describe patients

B. MRI scans and handwritten notes are unstructured data; the lab value spreadsheet is structured data

C. MRI scans are structured data; notes and lab values are unstruc-

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Question 14 | Domain 3 | Difficulty: Intermediate

Match each prompt engineering concept (1-3) to the example that best illustrates it (A-C).

Concepts:

1. Context
2. Instruction
3. Negative prompt

Examples:

A. "Do not include any pricing information in your response."

B. "Summarize the attached customer complaint in two sentences."

C. "The customer has been a member for five years and previously contacted support twice about

Match each numbered concept to its corresponding lettered example.

Question 15 | Domain 5 | Difficulty: Accessible

A company wants to automatically discover and classify any personally identifiable information (PII) that may have accidentally been stored in an S3 bucket used for AI training data. Which AWS service should they use?

A. Amazon Inspector B. Amazon Macie C. AWS PrivateLink D. AWS Key Management Service

Question 16 | Domain 2 | Difficulty: Accessible

A company wants to use an AI model that can accept both an image and a text question about that image, then generate a relevant text answer. Which type of model is best suited for this use case?

A. A diffusion model
B. A multimodal model
C. A single-modality clustering model
D. A traditional regression model

Question 17 | Domain 3 | Difficulty: Accessible

A company already uses Amazon RDS for PostgreSQL for its application data and wants to add vector storage for a RAG application without introducing an entirely new database technology. Which approach best fits this need?

A. Migrate to Amazon Neptune immediately

B. Use the pgvector extension within Amazon RDS for PostgreSQL to store vector embeddings

C. Store vector embeddings as plain text files in Amazon S3 Glacier

D. Use Ama-

Question 18 | Domain 1 | Difficulty: Intermediate Place the following stages of a typical AI/ML development pipeline into the correct order, from first to last.

1. Model training 2. Data collection 3. Deployment 4. Data preprocessing 5. Model evaluation *Arrange items 1-5 into the correct sequence.*

Question 19 | Domain 4 | Difficulty: Intermediate

When evaluating two foundation models with similar accuracy, a company's responsible AI policy requires considering the environmental footprint of training and running each model before making a final selection. Which

A. Fairness B. Environmental sustainability considerations in model selection C. Explainability D. Data lineage

Question 20 | Domain 2 | Difficulty: Intermediate

A company's chatbot application sends very long background instructions with every single user message to maintain consistent behavior, in addition to the model's lengthy response each time. The team notices their monthly Amazon Bedrock

A. Bedrock charges a flat monthly fee regardless of usage, so usage patterns are irrelevant

B. Both the long input instructions and the long output responses count toward billable tokens, so longer prompts and longer responses directly increase cost

C

Question 21 | Domain 3 | Difficulty: Intermediate

A company wants to teach a foundation model company-specific terminology and tone, has a modest budget, and does not want to retrain the entire model from scratch or build a permanent retrieval system. Which customization approach offers

A. Pre-training a new model from scratch
B. Fine-tuning the existing foundation model on company-specific examples
C. Doing nothing and using the base model as-is
D. Building a brand-new transformer architecture

Question 22 | Domain 5 | Difficulty: Accessible

An auditor wants to understand exactly where the training data for a deployed model came from and how it was transformed before being used. Which concept and AWS tool combination most directly supports this need?

A. Encryption in transit, using AWS KMS
B. Data lineage, documented using Amazon SageMaker Model Cards
C. Token-based pricing, tracked using AWS Budgets
D. Prompt engineering, managed using Amazon Bedrock Prompt Management

Question 23 | Domain 1 | Difficulty: Accessible

A logistics company wants to automatically convert audio recordings of customer service calls into written transcripts so that compliance staff can review them as text. Which AWS service is purpose-built for this task?

A. Amazon Polly
B. Amazon Transcribe
C. Amazon Comprehend
D. Amazon Translate

Question 24 | Domain 2 | Difficulty: Intermediate

A development team wants their AI agent to be able to securely connect to multiple external tools and data sources, such as a company calendar and a ticketing system, using a standardized way of describing available tools and exchang-

A. Model Context Protocol (MCP)
B. Token-based pricing
C. Diffusion modeling
D. Reinforcement learning from human feedback (RLHF)

Question 25 | Domain 3 | Difficulty: Challenging

A security team is reviewing risks associated with a customer-facing GenAI chatbot built using prompt engineering. Select TWO risks they should specifically evaluate for.

A. Prompt injection/poisoning, where malicious instructions are smuggled into the model via untrusted input data B. Excessive use of the temperature parameter set to exactly 0.5 C. Jailbreaking, where users attempt to bypass the model's safety re-

Select TWO.

Question 26 | Domain 4 | Difficulty: Challenging A company's legal team is assessing risks before launching a GenAI content-generation tool for public use. Select THREE legal risks they should evaluate.

A. Intellectual property infringement claims arising from training data or generated outputs
B. Loss of customer trust resulting from AI failures or embarrassing outputs
C. The number of GPUs used to train the

Select THREE.

Question 27 | Domain 2 | Difficulty: Challenging

A company is selecting a foundation model for a new customer-facing application. Select THREE factors that should reasonably influence the model selection decision.

A. The model's latency (how quickly it returns a response)

B. The model's compliance with relevant data-handling regulations

C. The favorite programming language of the lead developer

D. The cost per token or cost per inference

E. Th

Select THREE.

Question 28 | Domain 3 | Difficulty: Accessible

A company wants an AI system that can autonomously check inventory levels, place a reorder with a supplier's system, and send a confirmation email, all without a human manually performing each step. Which AI capability best

A. A simple classification model
B. An AI agent capable of multi-step autonomous task completion
C. A static rules-based spreadsheet macro
D. A single-shot prompt template

Question 29 | Domain 1 | Difficulty: Intermediate

A subscription streaming company has years of customer data with no predefined categories and wants to discover natural groupings of viewers with similar watching habits, without knowing in advance how many groups exist or what de-

A. Regression
B. Classification
C. Clustering
D. Reinforcement learning

Question 30 | Domain 5 | Difficulty: Intermediate

A healthcare AI team needs to use real patient records for model development while minimizing the risk of exposing identifiable individual information during that process. Which category of technique addresses this need?

A. Privacy-enhancing technologies, such as data masking or differential privacy
B. Increasing the model's context window
C. Switching to a multimodal model
D. Using a higher temperature setting

Question 31 | Domain 2 | Difficulty: Intermediate

A company's AI agent needs to remember a customer's preferences across multiple separate conversations occurring days apart, rather than treating each conversation as if it had never happened before. Which agentic AI concept addresses this

A. Workflow orchestration
B. Tool usage
C. Memory management
D. Tokenization

Question 32 | Domain 3 | Difficulty: Intermediate

A team wants the model to classify customer feedback into “positive,” “neutral,” or “negative” and provides the model with three example feedback messages, each already labeled with its correct category, before asking it to clas-

A. Zero-shot prompting
B. Few-shot prompting C. Chain-of-thought prompting
D. Negative prompting

Question 33 | Domain 4 | Difficulty: Accessible

A team building a facial recognition training dataset wants to ensure people of many different ages, ethnicities, and lighting conditions are well represented, rather than skewed toward one narrow group. Which responsible AI dataset character-

A. Tokenization
B. Inclusivity and balanced representation
C. Encryption at rest
D. Model distillation

Question 34 | Domain 1 | Difficulty: Intermediate

A financial services firm is deciding between a traditional ML model and a foundation model for a loan-approval decision system that must be explainable to regulators. Select TWO factors that would make a traditional ML model th

A. The decision must be easily explainable to a regulator
B. The application needs to generate creative marketing copy
C. The organization has strict requirements to trace exactly how each input feature affected the outcome
D.

Select TWO.

Question 35 | Domain 2 | Difficulty: Accessible

A company's customer support team is impressed that their new GenAI assistant can handle questions about returns, shipping, product specs, and account issues, all using natural conversational language, without needing a separate program built

A. Determinism
B. Adaptability across a wide range of tasks and domains using natural language
C. Guaranteed factual accuracy
D. Zero operating cost

Question 36 | Domain 3 | Difficulty: Accessible

A team's prompts are producing inconsistent, overly long, and sometimes irrelevant responses from their foundation model. Which prompt engineering best practice would most directly help address this issue?

A. Make the prompt as vague as possible to give the model maximum freedom
B. Write a clear, specific, and appropriately concise prompt with well-defined instructions
C. Remove all context from the prompt to reduce token usage
D

Question 37 | Domain 5 | Difficulty: Challenging

A company wants its GenAI application to anchor every factual claim it makes to a verified source document, reducing the chance that the model invents information not actually supported by company data. Which technique most directly

A. Increasing the temperature parameter B. RAG grounding C. Reducing the number of fine-tuning examples D. Disabling output filtering

Question 38 | Domain 2 | Difficulty: Intermediate

A company deployed a GenAI writing assistant for its marketing team six months ago. Leadership wants to know whether the investment is paying off in business terms. Which metric would most directly help answer that question?

A. The number of parameters in the underlying foundation model

B. Return on investment (ROI), comparing cost savings and revenue impact against the tool's cost

C. The model's context window size in tokens

D. The diffusion

Question 39 | Domain 3 | Difficulty: Challenging

A company is preparing a dataset to fine-tune a foundation model for internal customer service use. Select THREE practices that represent good data preparation for fine-tuning.

A. Curating high-quality, relevant examples rather than including every available record regardless of quality B. Ensuring the training data is properly licensed and legally usable C. Including only data from a single customer

Select THREE.

Question 40 | Domain 1 | Difficulty: Intermediate

A game development studio wants an AI agent that learns to play a new game by repeatedly attempting moves, receiving a positive score when it succeeds and a penalty when it fails, gradually improving its strategy over time. Which

A. Supervised learning B. Unsupervised learning C. Reinforcement learning D. Transfer learning

Question 41 | Domain 4 | Difficulty: Intermediate

A model performs extremely well on its training data but performs poorly on new, unseen data in production. What does this most likely indicate?

A. Underfitting B. Overfitting C. High bias with no variance D. Perfect generalization

Question 42 | Domain 2 | Difficulty: Challenging

A development team is deciding which AWS offerings to use for building and then operating an AI agent in production. Select TWO statements that correctly describe the distinction between Strands Agents and Amazon Bedrock AgentCore.

A. Strands Agents is an open-source framework used to build agents; AgentCore is the managed service used to deploy, secure, and scale agents in production B. AgentCore and Strands Agents are simply two different names for the exact same offering C. Strands

Select TWO.

Question 43 | Domain 3 | Difficulty: Intermediate

A company has multiple teams editing and experimenting with prompts for the same production GenAI application, and leadership is concerned about losing track of which prompt version is currently live versus which are in test-

A. Amazon SageMaker Clarify B. Amazon Bedrock Prompt Management C. AWS Audit Manager D. Amazon Macie

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Question 44 | Domain 5 | Difficulty: Challenging

A security team is building a checklist of AI-specific security and privacy considerations to address before launching a customer-facing GenAI application. Select THREE items that belong on this AI-specific checklist.

A. Prompt injection defenses, to prevent malicious instructions hidden in user input or external data
B. Output filtering and validation, to catch inappropriate or incorrect content before it reaches users
C. Toxicity de-

Select THREE.

Question 45 | Domain 1 | Difficulty: Challenging

A hospital is deploying an AI model to flag potentially cancerous scans for further review by a radiologist. Missing an actual cancer case is far more costly than sending a healthy scan for unnecessary extra review. Which

A. Precision
B. Recall
C. F1 score only, ignoring precision and recall individually
D. Accuracy

Question 46 | Domain 2 | Difficulty: Accessible

A small business with no in-house ML engineers wants to add a GenAI-powered chatbot to its website quickly and affordably. What is a key advantage of using a managed AWS GenAI service like Amazon Bedrock over building a foundation model from

A. Building from scratch is always cheaper for small businesses
B. Managed services provide lower barriers to entry, faster time to market, and reduced operational overhead compared to training a model from scratch
C. Managed services require

Question 47 | Domain 3 | Difficulty: Intermediate

A foundation model provider periodically updates its model by feeding it new, broad data over time to keep its general knowledge current, without targeting any single specific task or domain. Which training stage does this describe?

A. Fine-tuning
B. Continuous pre-training
C. Instruction tuning
D. Model distillation

Question 48 | Domain 4 | Difficulty: Challenging

An organization wants to both detect bias in a trained model and incorporate human review for predictions where the model has low confidence. Select TWO AWS tools that directly support these two distinct needs, respectively.

A. Amazon SageMaker Clarify, for detecting bias in models and datasets
B. Amazon Augmented AI (A2I), for routing low-confidence predictions to human reviewers
C. AWS CloudTrail, for detecting bias in datasets
D. Amazon Ma-

Select TWO.

Question 49 | Domain 2 | Difficulty: Challenging

A healthcare company is choosing between building its own GenAI infrastructure from scratch or using AWS-managed GenAI services, partly because it needs to meet strict healthcare compliance requirements. Which benefit of AWS infrastructure

A. AWS infrastructure includes built-in security certifications, compliance capabilities, and clearly defined responsibility boundaries that support regulated workloads

B. AWS infrastructure guarantees the company will never

Question 50 | Domain 3 | Difficulty: Challenging

A team is designing an evaluation strategy for a new customer-facing foundation model application before launch. Select TWO evaluation approaches that are documented as appropriate methods for assessing FM performance.

A. Human-in-the-loop evaluation, where people rate the quality of model outputs
B. Comparing outputs against benchmark datasets with known reference answers
C. Asking the original model itself, with no other validation, whether its own an-

Select TWO.

Question 51 | Domain 5 | Difficulty: Accessible

A company wants to download AWS's own existing compliance certifications and audit reports (such as SOC 2 reports) to share with its own auditors, without having to generate any new evidence itself. Which AWS service should they use?

A. AWS Audit Manager B. AWS Artifact C. AWS Config D. Amazon Inspector

Question 52 | Domain 1 | Difficulty: Challenging

A company is mapping its planned AI initiatives to AI application categories. Select the THREE initiatives that are correctly matched to their category.

A. Using AI to flag unusual credit card transactions in real time - fraud detection

B. Using AI to recommend products based on browsing history - recommendation system

C. Using AI to predict next quarter's re-

Select THREE.

Question 53 | Domain 2 | Difficulty: Intermediate

A company has a GenAI application with highly predictable, consistently high request volume throughout the day and wants guaranteed, consistent response times rather than variable performance during peak demand. Which Amazon Bedrock pric-

A. On-demand token-based pricing only B. Provisioned throughput C. Spot pricing for foundation models D. Free-tier access only

Question 54 | Domain 3 | Difficulty: Accessible

A company is evaluating an FM-powered tool that produces translated product descriptions from English into French and wants an automated metric specifically suited to assessing translation quality. Which metric is most appropriate?

A. ROUGE B. BLEU C. F1 score D. Accuracy

Question 55 | Domain 4 | Difficulty: Accessible

A company wants to produce structured documentation describing a model's intended use, performance characteristics, and known limitations, to share with stakeholders and auditors. Which AWS tool is designed for this purpose?

A. Amazon SageMaker Model Cards B. AWS Trusted Advisor C. Amazon Polly D. AWS Glue

Question 56 | Domain 2 | Difficulty: Accessible

A data scientist wants to quickly deploy a pre-built foundation model directly from within the SageMaker interface, as part of a broader custom ML workflow she's already building in SageMaker. Which service is most aligned with this need?

A. Amazon Bedrock AgentCore
B. Amazon SageMaker JumpStart
C. AWS Audit Manager
D. Amazon Macie

Question 57 | Domain 3 | Difficulty: Intermediate

A team has thousands of chatbot responses to evaluate for quality and tone but does not have the staffing to manually review each one. They want a scalable automated method that goes beyond simple word-overlap metrics. Which evalua-

A. ROUGE score only
B. LLM-as-a-judge, using a separate trusted model to evaluate the quality of the outputs
C. Manual review by the original engineering team of every single response
D. BLEU score only

Question 58 | Domain 1 | Difficulty: Intermediate

A retail demand-forecasting model was highly accurate when launched, but six months later its predictions have become noticeably less reliable as customer buying patterns shifted post-launch. According to MLOps best practices, what should the team do?

A. Decommission the model permanently, since AI models cannot adapt to changing conditions

B. Continue using the model unchanged, since retraining is only needed if the original code has bugs

C. Monitor the model's performance and

Question 59 | Domain 5 | Difficulty: Intermediate

A multinational company must ensure that AI training data for its European customers remains stored only within European AWS regions, due to local regulatory requirements. Which data governance concept does this requirement reflect?

A. Data lineage B. Data residency C. Data cataloging D. Token-based pricing

Question 60 | Domain 1 | Difficulty: Challenging

An AI practitioner is evaluating two candidate models for a customer support chatbot. Model A is 3% more accurate than Model B but costs four times as much per month to run, and the support team reports no noticeable

A. Model A should automatically be chosen because higher accuracy always means higher business value

B. Model B may be the better business choice, since the marginal accuracy gain from Model A does not appear to translate into better busi-

Question 61 | Domain 3 | Difficulty: Challenging

A company has deployed an AI agent that handles customer support tickets end-to-end. Select TWO metrics that would help determine whether the agent is meeting business objectives.

A. Task completion rate - the percentage of support tickets the agent successfully resolves without escalation B. Cost per interaction - the total cost to handle one support ticket using the agent C. The number of lines

Select TWO.

Question 62 | Domain 4 | Difficulty: Intermediate

A company deploys an AI tool that automatically flags certain loan applications for additional review. To support human-centered design principles, what should the company also provide to applicants whose loans were flagged?

A. No explanation at all, since the model's internal logic is proprietary

B. A way for applicants to understand that AI played a role in the decision and a mechanism to provide feedback or raise concerns

C. The complete

Question 63 | Domain 1 | Difficulty: Accessible

A company wants to add a feature to its mobile app that reads order confirmation text aloud to visually impaired customers in a natural-sounding voice. Which AWS service should they use?

A. Amazon Lex
B. Amazon Polly
C. Amazon Transcribe
D. Amazon Comprehend

Question 64 | Domain 5 | Difficulty: Challenging

A company wants to use a recognized framework to help classify its various GenAI use cases by the level of risk and responsibility involved, in order to apply appropriately scaled governance controls to each one. Which gov-

A. The AWS Well-Architected Framework
B. The Generative AI Security Scoping Matrix
C. The AWS Shared Responsibility Model
D. ROUGE scoring

Question 65 | Domain 3 | Difficulty: Accessible

A company has a very large, highly accurate foundation model but needs a smaller, faster, less expensive version to run on resource-constrained devices, while retaining as much of the original model's behavior as possible. Which technique best fits

A. Model distillation B. RAG C. Prompt engineering D. Data labeling

End of Part 2. Proceed to Part 3 for the complete answer key.

PART 3: COMPLETE ANSWER KEY WITH EXPLANATIONS

Q1 - Answer: Data selection -> Pre-training -> Fine-tuning -> Evaluation -> Deployment Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: FM lifecycle

Why this order is correct: The FM lifecycle begins with selecting the data that will be used to train the model, then pre-training the model on that broad dataset to build general capability, then fine-tuning it for more specific tasks, then

Why other orderings are wrong: Any order that places fine-tuning before pre-training is wrong because a model must first acquire general capability through pre-training before it can be fine-tuned for a specific task. Any order that places evaluation b

Q2 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.1 | Concept: Deep learning

Why B is correct: Deep learning involves neural networks with multiple interconnected layers that automatically learn patterns and features directly from raw data (here, images), without requiring a human to manually define what those features look like.

Why A is wrong: Robotic process automation (RPA) automates repetitive, rules-based tasks using predefined steps; it does not learn patterns from images. **Why C is wrong:** A rules-based expert system requires humans to manually define decision logic in advance, which is the opposite of automatically learning patterns from data. **Why D is wrong:** A business intelligence dashboard visualizes existing data for human interpretation; it does not learn from data on its own.

Q3 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.1 | Concept: FM selection criteria

Why B is correct: Multi-lingual support, latency, and cost are all explicitly documented FM selection criteria used to choose the most appropriate foundation model for a specific use case.

Why A is wrong: Labeling quality is a data preparation consideration relevant to fine-tuning, not a criterion for selecting among existing foundation models. **Why C is wrong:** RLHF is a specific fine-tuning technique used to incorporate human feedback into training, not a general FM selection criterion. **Why D is wrong:** Diffusion model architecture is a technical detail specific to certain image-generation models, not a general selection criterion like the ones described in the scenario.

Q4 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: Chunking and embedding

Why B is correct: Chunking is the process of splitting a document into smaller pieces, and embedding is the process of converting each piece into a numeric vector that represents its meaning - exactly the two-step process described.

Why A is wrong: Tokenization breaks text into sub-word units for the model to process internally, not for splitting documents into retrievable pieces; fine-tuning is a training process unrelated to converting text into numeric meaning representations. **Why C is wrong:** Prompt engineering is about designing inputs to guide a model's output; distillation is about creating smaller models from larger ones. Neither matches the described process. **Why D is wrong:** Context engineering is about designing what information goes into a model's context window; pre-training is the model's initial broad training phase. Neither matches the described process.

Q5 - Answer: A Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Veracity

Why A is correct: Veracity is the responsible AI feature concerned with producing truthful and accurate outputs, which directly addresses the concern about a model fabricating claims.

Why B is wrong: Inclusivity concerns designing AI to work well for diverse populations, not the factual accuracy of outputs. **Why C is wrong:** Robustness concerns reliable performance under varied conditions, not the truthfulness of generated content. **Why D is wrong:** Confidence scoring is a grounding/security technique that provides a measure of certainty about an output; it is not itself one of the six core responsible AI features (bias, fairness, inclusivity, robustness, safety, veracity).

Q6 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.1 | Concept: Temperature (inference parameter)

Why B is correct: Increasing the temperature setting makes model outputs more random and creative, which directly supports the goal of generating more varied and unconventional story ideas.

Why A is wrong: Decreasing temperature makes outputs more conservative and predictable, the opposite of what's needed for creative variety. **Why C is wrong:** Context window size determines how much input the model can consider; it does not control how creative or random the output is. **Why D is wrong:** Fine-tuning epochs relate to a training process, not to an inference-time parameter that adjusts output randomness during generation.

Q7 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.1 | Concept: Shared Responsibility Model

Why B is correct: Under the AWS Shared Responsibility Model, customers are responsible for "security in the cloud," which includes configuring IAM permissions and managing the data they send to and store within their applications.

Why A is wrong: Physical data center security is part of AWS's "security of the cloud" responsibility, not the customer's. **Why C is wrong:** Maintaining the underlying hardware that runs a managed service like Bedrock is AWS's responsibility. **Why D is wrong:** For a managed service like Bedrock, AWS is responsible for patching the host operating system of the underlying infrastructure.

Q8 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.1 | Concept: Batch inferencing

Why B is correct: Batch inferencing processes a large volume of data at once, typically on a schedule, when immediate individual responses are not required - exactly matching this overnight, large-volume use case.

Why A is wrong: Real-time inferencing returns a response within milliseconds to seconds for individual requests, which is not the requirement described. **Why C is wrong:** Asynchronous inferencing is designed for large individual payloads with delayed responses, but is not the standard choice for scheduled, very large batch jobs like this one. **Why D is wrong:** Serverless inferencing describes an infrastructure management approach, not the batch-versus-immediate distinction being tested here.

Q9 - Answer: A Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: Context window

Why A is correct: The context window is the amount of text, measured in tokens, that a model can consider at one time. Once a conversation grows beyond that limit, earlier content can no longer be "seen" by the model, producing exactly the behavior described.

Why B is wrong: Hallucination refers to the model generating confidently incorrect information, not to losing track of earlier conversation content due to length limits. **Why C is wrong:** Fine-tuning issues would generally affect model behavior across the board, not specifically cause forgetting tied to conversation length. **Why D is wrong:** Batch inferencing relates to processing large volumes of requests on a schedule and is unrelated to conversational memory limits.

Q10 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.1 | Concept: Retrieval Augmented Generation (RAG)

Why B is correct: RAG allows a foundation model to retrieve relevant information from a company's own knowledge base, such as case files, before generating a response, grounding the answer in that organization's specific data.

Why A is wrong: Model distillation creates a smaller, faster model trained on a larger model's outputs; it does not connect a model to a company's documents at query time. **Why C is wrong:** Prompt poisoning is a security risk involving maliciously injected instructions, not a legitimate grounding technique. **Why D is wrong:** Reinforcement learning is a training paradigm based on rewards, not a method for grounding responses in retrieved documents.

Q11 - Answer: B Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Amazon Bedrock Guardrails

Why B is correct: Amazon Bedrock Guardrails allows organizations to configure content filters and topic restrictions that prevent a model from discussing specific topics, exactly the capability needed here.

Why A is wrong: SageMaker Clarify detects bias in models and datasets; it does not enforce live topic restrictions on a deployed chatbot. **Why C is wrong:** CloudTrail logs API activity for auditing purposes; it does not restrict what a model can say. **Why D is wrong:** Amazon Inspector scans compute workloads for vulnerabilities and has no role in restricting chatbot topics.

Q12 - Answer: A, C Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.2 | Concept: GenAI disadvantages

Why A is correct: Hallucination is a well-documented GenAI risk in which the model confidently produces inaccurate information, a serious concern for unreviewed customer-facing content. **Why C is correct:** Nondeterminism means identical prompts can yield different outputs across runs, which complicates quality assurance for content that will not be reviewed by a human before sending.

Why B is wrong: Managed services such as Amazon Bedrock specifically remove the need to manually provision servers; this is not an inherent disadvantage of GenAI itself. **Why D is wrong:** Multi-modal GenAI models can generate text, images, audio, and video; this statement is factually incorrect. **Why E is wrong:** Understanding and responding to natural language is a core capability of GenAI/LLMs, not a limitation.

Q13 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.1 | Concept: Structured vs. unstructured data

Why B is correct: Structured data fits neatly into a predefined row-and-column format, like the lab value spreadsheet. MRI scans (images) and handwritten notes (free text) do not have this predefined structure and are therefore unstructured data.

Why A is wrong: Images and free text are unstructured, not structured, regardless of subject matter. **Why C is wrong:** This reverses the correct classification; MRI scans are unstructured (image data), and the spreadsheet is structured. **Why D is wrong:** Labeled/unlabeled describes whether data has target outcomes attached for supervised learning, which is a separate concept from data format (structured/unstructured); this answer conflates the two.

Q14 - Answer: 1-C, 2-B, 3-A Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.2 | Concept: Prompt engineering components

Why this matching is correct: Context (1) provides background information the model needs to respond appropriately, matching example C (the customer's membership history and prior contacts). Instruction (2) is the specific task the model is asked to perform, matching example B (summarize the complaint). Negative prompt (3) tells the model explicitly what to avoid including, matching example A (exclude pricing information).

Why other pairings are wrong: Pairing context with B or A would mistake background information for either a task instruction or an exclusion rule. Pairing instruction with C or A would mistake background information or an exclusion for an actionable task. Pairing negative prompt with B or C would mistake a task instruction or background context for guidance about what to avoid.

Q15 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.1 | Concept: Amazon Macie

Why B is correct: Amazon Macie is purpose-built to automatically discover and classify sensitive data, including PII, stored in Amazon S3.

Why A is wrong: Amazon Inspector scans compute workloads, such as EC2 instances, for vulnerabilities; it does not scan S3 content for sensitive data. **Why C is wrong:** AWS PrivateLink provides private network connectivity between VPCs and AWS services; it does not scan or classify data. **Why D is wrong:** AWS KMS manages encryption keys; it does not discover or classify sensitive content within stored files.

Q16 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: Multi-modal models

Why B is correct: Multi-modal models are specifically designed to accept and/or generate more than one type of data, such as both images and text, which is exactly what's needed to answer a text question about an image.

Why A is wrong: Diffusion models are primarily used to generate content, especially images, from noise; they are not designed to answer text questions about an existing image. **Why C is wrong:** Clustering models group unlabeled data and are unrelated to processing images and generating text answers. **Why D is wrong:** Regression models predict continuous numeric values and have no relevance to interpreting images and answering questions about them.

Q17 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.1 | Concept: Vector databases for RAG

Why B is correct: Amazon RDS for PostgreSQL supports the pgvector extension, which allows vector embeddings to be stored and queried within the same relational database technology the company already uses, avoiding the need for a new database platform.

Why A is wrong: Migrating to Amazon Neptune would introduce a new database technology, exactly what the company wants to avoid. **Why C is wrong:** S3 Glacier is archival storage optimized for infrequent access and is unsuited to the fast vector similarity queries RAG applications require. **Why D is wrong:** Amazon Personalize is a recommendation service, not a vector database, and is not designed to store embeddings for RAG.

Q18 - Answer: Data collection -> Data preprocessing -> Model training -> Model evaluation -> Deployment Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.3 | Concept: AI/ML pipeline

Why this order is correct: Raw data must be collected before it can be cleaned and transformed (preprocessing); a model can only be trained once data has been prepared; performance must be evaluated before a model is trusted in production; and deployment occurs only after the model has demonstrated it meets requirements.

Why other orderings are wrong: Placing model training before data preprocessing is wrong because models are trained on cleaned, prepared data, not raw data. Placing deployment before evaluation is wrong because organizations should validate performance before releasing a model into production. Placing data preprocessing before data collection is wrong because there is nothing to preprocess until data has first been gathered.

Q19 - Answer: B Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Environmental sustainability in model selection

Why B is correct: Responsible practices for model selection explicitly include considering the environmental impact, energy consumption, and sustainability implications of training and deployment choices, exactly the consideration described.

Why A is wrong: Fairness concerns equal treatment of groups affected by model outputs or decisions, not environmental footprint. **Why C is wrong:** Explainability concerns understanding why a model produces a given output, unrelated to environmental impact. **Why D is wrong:** Data lineage tracks where data came from and how it was transformed; it is unrelated to environmental sustainability of model training.

Q20 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: Token-based pricing

Why B is correct: Token-based pricing typically bills for both input tokens (including any repeated background instructions) and output tokens (the response); longer prompts and longer responses both directly increase total token consumption and cost.

Why A is wrong: Foundation model usage on Bedrock is generally billed based on tokens consumed (or provisioned throughput reserved), not a single flat fee unrelated to usage. **Why C is wrong:** Most token-based pricing models bill for both input and output tokens, not output tokens alone. **Why D is wrong:** Token-based pricing is a core characteristic of text-based FM usage broadly, not something limited to image generation models.

Q21 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.1 | Concept: FM customization cost tradeoffs

Why B is correct: Fine-tuning adapts an existing foundation model to specific terminology and tone using a smaller, targeted dataset, at substantially lower cost than pre-training a new model from scratch, making it the appropriate middle-ground option here.

Why A is wrong: Pre-training a new model from scratch is the most expensive and resource-intensive customization option, not a moderate-cost middle ground. **Why C is wrong:** Using the base model as-is would not achieve the stated goal of teaching it company-specific terminology and tone. **Why D is wrong:** Building an entirely new transformer architecture is far beyond the scope and cost level implied by the scenario and is not a standard FM customization approach covered on this exam.

Q22 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.1 | Concept: Data lineage and Model Cards

Why B is correct: Data lineage is the concept of tracking where data came from and how it was transformed, and Amazon SageMaker Model Cards can be used to document model provenance and training data sources, exactly addressing this auditor's need.

Why A is wrong: Encryption in transit protects data as it moves across networks; it does not document where data originated or how it was transformed. **Why C is wrong:** Token-based pricing and AWS Budgets relate to cost management, not to documenting data origin. **Why D is wrong:** Prompt engineering and Bedrock Prompt Management relate to designing and versioning prompts, not to documenting training data provenance.

Q23 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.2 | Concept: Amazon Transcribe

Why B is correct: Amazon Transcribe is an automatic speech recognition (ASR) service specifically designed to convert spoken audio into written text, exactly matching this requirement.

Why A is wrong: Amazon Polly converts text into speech, the opposite direction of what's needed. **Why C is wrong:** Amazon Comprehend analyzes the meaning of text that already exists in text form; it does not convert audio to text. **Why D is wrong:** Amazon Translate converts text from one language to another; it does not process audio input.

Q24 - Answer: A Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: Model Context Protocol (MCP)

Why A is correct: The Model Context Protocol (MCP) provides a standardized way for AI agents to connect to and interact with external tools and systems, exactly matching the described need.

Why B is wrong: Token-based pricing relates to billing for FM usage, not to connecting agents with external tools. **Why C is wrong:** Diffusion modeling is an image-generation technique, unrelated to agent-to-tool connectivity. **Why D is wrong:** RLHF is a fine-tuning technique that incorporates human feedback into model training; it is unrelated to standardized tool connectivity.

Q25 - Answer: A, C Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.2 | Concept: Prompt engineering risks

Why A is correct: Prompt injection/poisoning is a documented risk in which malicious instructions are embedded in untrusted data the model processes, potentially hijacking its intended behavior. **Why C is correct:** Jailbreaking is a documented risk in which users craft prompts specifically designed to bypass a model's built-in safety restrictions.

Why B is wrong: A specific temperature value is a normal inference parameter choice and is not inherently a security risk. **Why D is wrong:** Response length is a quality or cost consideration, not a documented security risk of prompt engineering. **Why E is wrong:** Font size is a UI/UX design issue entirely unrelated to prompt engineering security risks.

Q26 - Answer: A, B, D Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Legal risks of GenAI

Why A is correct: Intellectual property infringement is an explicitly documented legal risk associated with GenAI training data and generated outputs. **Why B is correct:** Loss of customer trust resulting from AI failures is an explicitly documented legal/business risk of GenAI deployment. **Why D is correct:** Hallucination-driven misinformation is an explicitly documented legal risk that can mislead end users and create liability for the organization.

Why C is wrong: GPU count is an infrastructure detail with no direct legal risk implication. **Why E is wrong:** Programming language choice is a technical implementation detail, not a legal risk factor.

Q27 - Answer: A, B, D Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.2 | Concept: GenAI model selection factors

Why A is correct: Latency directly affects the user experience and is an explicitly documented FM selection criterion. **Why B is correct:** Compliance with relevant regulations is an explicitly documented factor, particularly important for customer-facing or regulated applications. **Why D is correct:** Cost is explicitly listed as a key selection factor, since pricing can vary significantly between models.

Why C is wrong: A developer's preferred programming language is a development convenience, not a documented or relevant FM selection criterion. **Why E is wrong:** A vendor's social media following has no bearing on a model's technical suitability, compliance posture, or cost, and is not a documented selection factor.

Q28 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.1 | Concept: AI agents

Why B is correct: AI agents are specifically designed to autonomously complete multi-step tasks by taking sequences of actions, such as checking inventory, placing an order, and sending a confirmation, exactly matching this scenario.

Why A is wrong: A classification model only assigns a category to a given input; it does not autonomously perform a sequence of actions across multiple systems. **Why C is wrong:** A static macro can automate fixed, predefined steps but lacks the autonomous decision-making and adaptability that defines an AI agent. **Why D is wrong:** A single-shot prompt template is a prompt engineering technique for generating a single response, not an autonomous multi-step task-completion system.

Q29 - Answer: C Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.2 | Concept: Clustering

Why C is correct: Clustering is an unsupervised technique used to discover natural groupings in data when there are no predefined labels or categories, exactly matching this scenario.

Why A is wrong: Regression predicts a continuous numeric value, not groupings of similar items. **Why B is wrong:** Classification assigns data to predefined categories, but in this scenario the categories are not known in advance. **Why D is wrong:** Reinforcement learning involves an agent learning through trial-and-error rewards; it is not used to group existing customer data.

Q30 - Answer: A Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.1 | Concept: Privacy-enhancing technologies

Why A is correct: Privacy-enhancing technologies, such as data masking and differential privacy, are specifically designed to reduce the risk of exposing identifiable information while still allowing the data to be used for development.

Why B is wrong: Context window size affects how much text a model can process at once and has no relationship to protecting patient privacy. **Why C is wrong:** Multi-modal model selection concerns the types of data a model can process; it is unrelated to privacy protection. **Why D is**

wrong: Temperature is an inference parameter controlling output randomness and has no relevance to data privacy.

Q31 - Answer: C Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.1 | Concept: Memory management (agentic AI)

Why C is correct: Memory management in agentic AI refers to strategies for retaining and recalling relevant information across separate interactions, exactly what's needed to remember a customer's preferences across multiple conversations.

Why A is wrong: Workflow orchestration concerns coordinating the steps of a multi-step process, not retaining information across separate sessions over time. **Why B is wrong:** Tool usage refers to an agent's ability to call external tools to complete tasks, not to remembering customer information between conversations. **Why D is wrong:** Tokenization is the process of breaking text into processable units for the model; it is unrelated to retaining customer context across conversations.

Q32 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.2 | Concept: Few-shot prompting

Why B is correct: Few-shot prompting provides the model with multiple labeled examples, here three, before asking it to perform the same task on new input.

Why A is wrong: Zero-shot prompting provides no examples at all, which contradicts the scenario's description of three labeled examples. **Why C is wrong:** Chain-of-thought prompting guides the model to reason step by step; it is not specifically about providing labeled examples. **Why D is wrong:** Negative prompting tells the model what to avoid; it does not describe providing labeled examples to guide classification.

Q33 - Answer: B Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Dataset characteristics for responsible AI

Why B is correct: Ensuring diverse representation across ages, ethnicities, and conditions reflects the responsible AI dataset characteristics of inclusivity and balanced representation.

Why A is wrong: Tokenization is a text-processing technique unrelated to demographic representation in a dataset. **Why C is wrong:** Encryption at rest is a security control protecting stored data; it has no relationship to demographic balance. **Why D is wrong:** Model distillation is a model-size reduction technique unrelated to dataset composition or representativeness.

Q34 - Answer: A, C Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.2 | Concept: Traditional ML vs. foundation models

Why A is correct: Traditional ML models tend to be more interpretable than large foundation models, which is critical when a decision must be easily explained to a regulator. **Why C is correct:** A strict requirement to trace exactly how each input feature affected an outcome favors simpler, more interpretable traditional ML models over large, less transparent foundation models.

Why B is wrong: Generating creative content, such as marketing copy, is a strength of generative AI/foundation models, not a reason to favor traditional ML. **Why D is wrong:** Handling many varied, unrelated tasks without retraining is a strength of foundation models, not traditional ML, so this would argue for an FM, not against one. **Why E is wrong:** Traditional supervised ML actually requires labeled historical data to train; lacking such data argues against using traditional ML, not for it.

Q35 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.2 | Concept: GenAI advantages

Why B is correct: GenAI's adaptability allows a single model to flexibly handle many different topics and tasks through natural language, without requiring separate dedicated software built for each scenario, exactly what's demonstrated here.

Why A is wrong: GenAI is actually known for nondeterminism (variable outputs across runs), not determinism; this is the opposite of an accurate description of GenAI behavior. **Why C is wrong:** GenAI does not guarantee factual accuracy and is prone to hallucination; this is a documented limitation, not an advantage demonstrated in the scenario. **Why D is wrong:** GenAI services have real operating costs, such as token-based pricing; "zero cost" is not an accurate characterization.

Q36 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.2 | Concept: Prompt engineering best practices

Why B is correct: Specificity and appropriate concision are documented prompt engineering best practices that directly improve response quality and consistency, addressing the issues described.

Why A is wrong: Vague prompts tend to produce inconsistent and less relevant results, the opposite of what's needed to fix this problem. **Why C is wrong:** Removing necessary context can reduce response relevance and quality, even though it may reduce token usage. **Why D is wrong:** Systematic experimentation and revision of prompts is a recommended best practice; avoiding revision works directly against improving results.

Q37 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.1 | Concept: RAG grounding

Why B is correct: RAG grounding anchors model responses in retrieved, verified source documents, directly reducing the risk of the model inventing claims not supported by company data.

Why A is wrong: Increasing temperature makes output more random and creative, which would likely increase, not decrease, the risk of unsupported claims. **Why C is wrong:** Reducing the number of fine-tuning examples does not anchor responses to verified sources; it could simply reduce the model's task-specific knowledge without addressing grounding. **Why D is wrong:** Disabling output filtering removes a safety control rather than adding a grounding mechanism, and would not help anchor claims to verified sources.

Q38 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.2 | Concept: Business value metrics for GenAI

Why B is correct: ROI directly measures whether the financial benefits of a GenAI investment, such as efficiency gains or revenue impact, outweigh its costs, exactly what leadership is asking about.

Why A is wrong: Parameter count is a technical architecture detail, not a business value metric. **Why C is wrong:** Context window size is a technical capability metric, unrelated to measuring financial payoff. **Why D is wrong:** Image resolution settings are irrelevant to a text-based writing assistant and do not measure business value.

Q39 - Answer: A, B, D Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.3 | Concept: Data preparation for fine-tuning

Why A is correct: Data curation, selecting relevant, high-quality examples rather than every available record, is a documented best practice for fine-tuning data preparation. **Why B is correct:** Governance, ensuring data is legally usable and properly licensed, is an explicitly documented data preparation consideration for fine-tuning. **Why D is correct:** Sizing and representativeness, ensuring the dataset is large enough and reflects real-world inputs, are documented data preparation requirements.

Why C is wrong: Using data from only one customer segment when the model must serve all segments equally would produce an unrepresentative, biased dataset, directly contradicting the representativeness principle. **Why E is wrong:** Accurate labeling is a documented requirement for fine-tuning approaches that rely on labeled examples; skipping it undermines the fine-tuning process when labels are needed.

Q40 - Answer: C Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.1 | Concept: Reinforcement learning

Why C is correct: Reinforcement learning involves an agent learning through trial and error, guided by rewards and penalties, exactly as described in the scenario.

Why A is wrong: Supervised learning requires labeled input-output pairs provided in advance, not a reward signal generated through repeated trial and error. **Why B is wrong:** Unsupervised learning finds patterns in unlabeled data without any reward or feedback signal at all. **Why D is wrong:** Transfer learning applies knowledge learned in one context to another; it is a fine-tuning approach, not a learning paradigm built around rewards.

Q41 - Answer: B Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Overfitting

Why B is correct: Overfitting occurs when a model learns the training data, including its noise, too closely, resulting in excellent training performance but poor generalization to new, unseen data, exactly the pattern described.

Why A is wrong: Underfitting describes a model that performs poorly even on its own training data because it hasn't learned the underlying patterns well enough; this scenario shows excellent training performance, the opposite situation. **Why C is wrong:** High bias typically corresponds with underfitting, not the scenario described, which instead suggests high variance. **Why D is wrong:** Poor performance on new, unseen data is the opposite of perfect generalization.

Q42 - Answer: A, D Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.3 | Concept: Strands Agents vs. Amazon Bedrock AgentCore

Why A is correct: This is the documented distinction: Strands Agents is the open-source framework used to build agents, while AgentCore is the managed service used to deploy, secure, and scale those agents in production. **Why D is correct:** AgentCore's role is to provide production-grade capabilities, such as security and scaling, for agents that may have been built using frameworks like Strands Agents.

Why B is wrong: They are distinct offerings with different purposes (development framework versus managed production runtime), not the same offering under two names. **Why C is wrong:** There is no strict requirement that Strands Agents must always be used as a prerequisite for AgentCore; agents can be developed using various approaches. **Why E is wrong:** Strands Agents is a development framework for building agents, not a database service for storing logs.

Q43 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.2 | Concept: Amazon Bedrock Prompt Management

Why B is correct: Amazon Bedrock Prompt Management supports prompt versioning, helping teams track, organize, and control different prompt versions across testing and production, directly addressing this concern.

Why A is wrong: SageMaker Clarify is used for bias detection and explainability, not prompt versioning. **Why C is wrong:** AWS Audit Manager assists with compliance evidence collection, unrelated to managing prompt versions. **Why D is wrong:** Amazon Macie detects sensitive data in S3 and has no role in prompt version control.

Q44 - Answer: A, B, C Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.1 | Concept: AI-specific security and privacy considerations

Why A is correct: Prompt injection is an explicitly documented AI-specific security risk that an organization needs to defend against before launch. **Why B is correct:** Output filtering and validation are explicitly documented security and privacy considerations for AI systems. **Why C is correct:** Toxicity detection in AI outputs is explicitly documented as a security and privacy consideration for AI systems.

Why D is wrong: Font choice is a branding/design decision with no security or privacy relevance. **Why E is wrong:** Color palette selection is a UI design choice unrelated to AI-specific security and privacy considerations.

Q45 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.3 | Concept: Recall

Why B is correct: Recall measures the proportion of actual positive cases (cancer present) the model correctly identifies. Prioritizing recall minimizes missed cancer cases, even at the cost of more false alarms, matching the scenario's stated priorities.

Why A is wrong: Precision measures how many flagged cases are truly positive; optimizing for precision alone risks missing real cancer cases that aren't flagged, which is the costlier error described in the scenario. **Why C is wrong:** F1 score balances precision and recall equally and doesn't reflect the scenario's explicit preference for minimizing missed cases over minimizing false alarms. **Why D is wrong:** Accuracy can be misleading with imbalanced data (most scans are healthy) and does not address the asymmetric cost of a missed cancer case.

Q46 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.3 | Concept: Benefits of AWS GenAI services

Why B is correct: This reflects the documented advantages of AWS GenAI services: accessibility, lower barrier to entry, faster time to market, and cost-effectiveness compared to building and training a foundation model from scratch.

Why A is wrong: Training a foundation model from scratch is typically far more expensive and resource-intensive than using a managed service, especially for a small business with no in-house ML team. **Why C is wrong:** A core benefit of managed services is that AWS handles the underlying infrastructure, not the customer. **Why D is wrong:** Managed services like Bedrock are specifically designed to provide API access to existing foundation models, contradicting this statement.

Q47 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.3 | Concept: Continuous pre-training

Why B is correct: Continuous pre-training involves ongoing updates to a model using new, broad data over time, without targeting one specific task, exactly matching this description.

Why A is wrong: Fine-tuning adapts a model to a specific task or domain using a smaller, targeted dataset, not broad ongoing general updates. **Why C is wrong:** Instruction tuning is a fine-tuning method that trains on task-instruction pairs, which is more targeted than the broad updates described in the scenario. **Why D is wrong:** Model distillation compresses knowledge from a large model into a smaller one; it is unrelated to ongoing general knowledge updates.

Q48 - Answer: A, B Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.1 | Concept: Bias detection and human review tools

Why A is correct: SageMaker Clarify is explicitly designed to detect bias in trained models and datasets. **Why B is correct:** Amazon A2I is explicitly designed to add human review workflows specifically for low-confidence model predictions.

Why C is wrong: CloudTrail logs account API activity for auditing purposes; it does not detect statistical bias in datasets or models. **Why D is wrong:** Amazon Macie discovers sensitive data in S3; it has no human-review workflow capability. **Why E is wrong:** S3 Glacier is archival storage and is entirely unrelated to human review workflows.

Q49 - Answer: A Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.3 | Concept: AWS infrastructure benefits for GenAI

Why A is correct: AWS provides security certifications, built-in compliance capabilities, and a clear shared responsibility model, all of which directly support customers operating in regulated industries like healthcare.

Why B is wrong: The shared responsibility model means the customer still retains compliance obligations for what they build on top of AWS; AWS does not eliminate all customer compliance work. **Why C is wrong:** Encryption remains an important practice the customer should implement and configure appropriately; using AWS does not make encryption unnecessary. **Why D is wrong:** Using AWS infrastructure does not automatically certify a specific application as HIPAA compliant; the customer must still build and configure their application appropriately to meet such requirements.

Q50 - Answer: A, B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.4 | Concept: FM evaluation approaches

Why A is correct: Human-in-the-loop evaluation, having people rate model outputs, is an explicitly documented FM evaluation approach. **Why B is correct:** Comparing outputs to benchmark datasets with known reference answers is an explicitly documented evaluation approach.

Why C is wrong: While LLM-as-a-judge uses a separate, trusted model to evaluate outputs, relying solely on the same model's unverified self-assessment as the sole metric, with no other validation, is not the documented, reliable evaluation approach covered on this exam. **Why D is wrong:** GPU usage during training is an infrastructure/cost detail, not a method for evaluating output performance. **Why E is wrong:** Parameter count is an architectural detail and does not measure how well a model performs on actual tasks.

Q51 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.2 | Concept: AWS Artifact

Why B is correct: AWS Artifact is the self-service portal where customers can download AWS's own existing compliance reports and certifications on demand, at no cost, exactly matching this need.

Why A is wrong: AWS Audit Manager helps customers generate and collect their own ongoing compliance evidence; it is not where customers download AWS's pre-existing reports. **Why C is wrong:** AWS Config tracks resource configuration compliance over time; it does not provide AWS's own compliance certifications. **Why D is wrong:** Amazon Inspector scans for vulnerabilities; it does not provide compliance reports.

Q52 - Answer: A, B, C Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.2 | Concept: Real-world AI application categories

Why A is correct: Real-time anomaly detection on financial transactions is a textbook fraud detection use case. **Why B is correct:** Personalized product suggestions based on user behavior is the definition of a recommendation system. **Why C is correct:** Predicting future numeric business outcomes from historical trends is the definition of forecasting.

Why D is wrong: Converting spoken audio in one language to text in another blends speech recognition and translation, not computer vision, which involves analyzing visual images or video. **Why E is wrong:** Detecting and counting objects in video footage is computer vision, not natural language processing, which processes language and text.

Q53 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.3 | Concept: Provisioned throughput

Why B is correct: Provisioned throughput allows customers to reserve dedicated model capacity for consistent, predictable performance, exactly suited to a steady, high-volume workload that needs guaranteed throughput.

Why A is wrong: On-demand token-based pricing is flexible and scales with usage but does not guarantee dedicated capacity or consistent throughput during peak demand. **Why C is wrong:** Foundation model access through Bedrock is not offered through a spot-pricing model the way EC2 Spot Instances are. **Why D is wrong:** Free-tier access, where available, would not provide the guaranteed capacity required for consistently high production volumes.

Q54 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.4 | Concept: BLEU score

Why B is correct: BLEU (Bilingual Evaluation Understudy) is specifically designed to assess translation quality by comparing machine-generated translations to reference translations.

Why A is wrong: ROUGE is designed for evaluating summarization quality, not translation. **Why C is wrong:** F1 score is a classification metric balancing precision and recall; it is not designed for comparing translation output against reference text. **Why D is wrong:** Accuracy is a general classification metric and does not capture the nuances of translation quality the way BLEU does.

Q55 - Answer: A Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.2 | Concept: Amazon SageMaker Model Cards

Why A is correct: Amazon SageMaker Model Cards provide structured documentation of a model's intended use, performance characteristics, and known limitations, exactly matching this need.

Why B is wrong: AWS Trusted Advisor provides recommendations across cost, security, and performance for AWS accounts generally; it does not document model behavior. **Why C is wrong:** Amazon Polly is a text-to-speech service, unrelated to model documentation. **Why D is wrong:** AWS Glue is a data integration/ETL service, unrelated to model documentation.

Q56 - Answer: B Domain: Domain 2 - Fundamentals of GenAI | Task Statement: 2.3 | Concept: Amazon SageMaker JumpStart

Why B is correct: SageMaker JumpStart provides a curated library of pre-built models, including select foundation models, deployable with minimal setup directly within the SageMaker environment, exactly matching this need.

Why A is wrong: AgentCore is for deploying and securing production AI agents, not for quickly deploying pre-built models within SageMaker. **Why C is wrong:** AWS Audit Manager is a compliance evidence-collection service, unrelated to deploying foundation models. **Why D is wrong:** Amazon Macie is a data security service for detecting sensitive data in S3, unrelated to model deployment.

Q57 - Answer: B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.4 | Concept: LLM-as-a-judge

Why B is correct: LLM-as-a-judge uses a second, trusted LLM to evaluate the quality of outputs at scale, providing more nuanced assessment than simple overlap-based metrics, without requiring manual review of every response.

Why A is wrong: ROUGE measures word/phrase overlap with reference text and does not capture nuanced qualities like tone at the scale needed here. **Why C is wrong:** Manual review of every response was explicitly ruled out due to staffing constraints in the scenario. **Why D is wrong:** BLEU, like ROUGE, is designed around overlap-based comparison for translation tasks rather than nuanced judgment of quality and tone.

Q58 - Answer: C Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.3 | Concept: MLOps - model monitoring and retraining

Why C is correct: MLOps emphasizes ongoing model monitoring to detect performance degradation (model drift) and retraining with current data to keep the model accurate as real-world conditions change, exactly the recommended response.

Why A is wrong: Models can be retrained and improved; permanent decommissioning is an overreaction not supported by MLOps practice. **Why B is wrong:** Performance degradation due to shifting real-world patterns (data drift), not just code bugs, is a primary documented reason for retraining. **Why D is wrong:** Switching to a rules-based system abandons the value of ML rather than addressing the actual issue, which is outdated training data.

Q59 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.2 | Concept: Data residency

Why B is correct: Data residency refers to requirements that keep data within specified geographic regions, exactly matching this scenario's regulatory constraint.

Why A is wrong: Data lineage tracks where data came from and how it was transformed, not where it must physically reside. **Why C is wrong:** Data cataloging maintains a structured inventory of datasets; it does not address geographic storage requirements. **Why D is wrong:** Token-based pricing relates to GenAI cost models and is unrelated to geographic data storage requirements.

Q60 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.3 | Concept: Business metrics and ROI

Why B is correct: Business value depends on weighing technical performance against cost and actual outcome impact. A small accuracy gain that doesn't improve a metric that matters to the business (customer satisfaction) may not justify a fourfold cost increase.

Why A is wrong: Higher accuracy does not automatically equal higher business value if it doesn't improve outcomes that matter to the business and comes at a large cost premium. **Why C is wrong:** Cost is one of the standard business metrics (cost per user, ROI) used precisely for these kinds of comparisons; ignoring it is not appropriate. **Why D is wrong:** Hyperparameter tuning is a mathematical/technical activity that is out of scope for this exam and is not required to make a reasonable business-level comparison using metrics already available.

Q61 - Answer: A, B Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.4 | Concept: Business objective alignment metrics

Why A is correct: Task completion rate is an explicitly documented business objective alignment metric, directly measuring whether the agent achieves user goals. **Why B is correct:** Cost per interaction is an explicitly documented metric for understanding the business value and efficiency of an AI application.

Why C is wrong: Lines of code is a software development detail with no direct bearing on whether business objectives are being met. **Why D is wrong:** Parameter count is a technical architecture detail, not a business outcome metric. **Why E is wrong:** Interface color scheme is a design choice unrelated to measuring business objective alignment.

Q62 - Answer: B Domain: Domain 4 - Guidelines for Responsible AI | Task Statement: 4.2 | Concept: Human-centered design and AI decision transparency

Why B is correct: Human-centered design for explainable AI includes ensuring people understand the role AI played in a decision and providing feedback mechanisms for raising concerns, exactly what's described.

Why A is wrong: Providing no explanation at all contradicts the principle of AI decision transparency. **Why C is wrong:** Sharing complete proprietary source code is not what human-centered design principles call for; meaningful, understandable explanation is the goal, not raw code. **Why D is wrong:** Human-centered design does not promise the elimination of all future human review; in fact, human review (such as through A2I) is often a recommended responsible AI practice.

Q63 - Answer: B Domain: Domain 1 - Fundamentals of AI and ML | Task Statement: 1.2 | Concept: Amazon Polly

Why B is correct: Amazon Polly converts written text into natural-sounding spoken audio (text-to-speech), exactly matching this need.

Why A is wrong: Amazon Lex is for building conversational chatbots/voice bots that understand spoken or typed input, not simply reading text aloud. **Why C is wrong:** Amazon Transcribe converts speech into text, the opposite of what's needed here. **Why D is wrong:** Amazon Comprehend analyzes text for sentiment and entities; it does not produce audio output.

Q64 - Answer: B Domain: Domain 5 - Security, Compliance, and Governance for AI Solutions | Task Statement: 5.2 | Concept: Generative AI Security Scoping Matrix

Why B is correct: The Generative AI Security Scoping Matrix is a recognized governance framework specifically designed to help organizations classify GenAI use cases by risk level and apply appropriately scaled governance and security controls.

Why A is wrong: The AWS Well-Architected Framework evaluates workloads against six general best-practice pillars; it is not specific to scoping GenAI risk levels. **Why C is wrong:** The AWS Shared Responsibility Model clarifies the general division of security responsibilities between AWS and customers; it is not a use-case risk-scoping framework. **Why D is wrong:** ROUGE is an automated text-evaluation metric, unrelated to governance frameworks.

Q65 - Answer: A Domain: Domain 3 - Applications of Foundation Models | Task Statement: 3.3 | Concept: Model distillation

Why A is correct: Model distillation creates a smaller, faster model trained to mimic the outputs of a larger model, exactly the documented purpose of this technique and matching the stated goal.

Why B is wrong: RAG grounds responses in retrieved data; it does not create a smaller version of a model. **Why C is wrong:** Prompt engineering designs inputs to guide model behavior; it does not change the size or speed of the underlying model. **Why D is wrong:** Data labeling prepares training data with annotations; it does not directly produce a smaller, faster model.

End of Part 3.

PART 4: SCORING GUIDE

How to Score Your Exam

1. Count the number of questions you answered correctly out of 65.
2. For multiple-response questions, you must have selected every correct option and no incorrect options to count that question as correct. There is no partial credit.
3. For the ordering question and the matching question, every item must be in the correct position, or every pair must be correctly matched, to count as correct.
4. Divide your correct count by 65 to get your percentage score.
5. Use the table below to interpret your results.

Scoring Table

Score Range	Percentage	Interpretation	Recommendation
58-65 correct	89-100%	Exam Ready	Schedule your real exam within 2 weeks
52-57 correct	80-88%	Nearly Ready	Spend 1 more week reviewing weak areas, then retake this practice exam
46-51 correct	71-79%	On Track	Spend 2 more weeks studying, with a focus on your weakest domains
39-45 correct	60-70%	Needs Work	Retake the domain study modules covering your missed questions before attempting another practice exam
Below 39 correct	Below 60%	Not Ready	Return to the domain study modules and Master Glossary before retaking this practice exam

Important context on passing scores: The real AIF-C01 exam is scored on a scaled basis from 100 to 1,000, with a minimum passing score of 700. That scaled score is not the same as a simple percentage of questions answered correctly, since the real exam uses statistically weighted scoring across exam forms. For the purposes of this practice exam, treat **47 out of 65 (73%)** as your practice passing threshold. This is set slightly above the real exam's approximate percentage equivalent, to build in a safety margin since practice exam performance does not perfectly predict real exam performance.

PART 5: DOMAIN SCORE TRACKER

Because this practice exam mixes domains throughout, rather than grouping them, use the question number lists below to identify which questions belong to which domain. Fill in this tracker after completing the exam and checking your answers against Part 3.

Domain	Question Numbers	Total Questions	# Correct	Score %	Status
Domain 1: Fundamentals of AI and ML	2, 8, 13, 18, 23, 29, 34, 40, 45, 52, 58, 60, 63	13	_____	_____ %	_____

Domain	Question Numbers	Total Questions	# Correct	Score %	Status
Domain 2: Fundamentals of GenAI	1, 4, 9, 12, 16, 20, 24, 27, 31, 35, 38, 42, 46, 49, 53, 56	16	_____	_____ %	_____
Domain 3: Applications of Foundation Models	3, 6, 10, 14, 17, 21, 25, 28, 32, 36, 39, 43, 47, 50, 54, 57, 61, 65	18	_____	_____ %	_____
Domain 4: Guidelines for Responsible AI	5, 11, 19, 26, 33, 41, 48, 55, 62	9	_____	_____ %	_____
Domain 5: Security, Compliance, and Governance for AI Solutions	7, 15, 22, 30, 37, 44, 51, 59, 64	9	_____	_____ %	_____
Total	**All 65 questions**	**65**	_____	_____ %	_____

How to use this tracker: For the "Status" column, write "Strong" if your domain score is 80% or higher, "Adequate" if it is 70-79%, and "Needs Review" if it is below 70%. Any domain marked "Needs Review" should be the focus of your remediation plan in Part 6, regardless of your overall exam score, since a single very weak domain can be masked by strong performance elsewhere.

PART 6: REMEDIATION PLAN BY DOMAIN

Domain 1: Fundamentals of AI and ML

If you scored below 70% on Domain 1 questions, here is exactly what to review:

- Return to Task 3 (Domain 1 Study Module) and re-read the sections on AI/ML/deep learning distinctions, the four types of inferencing (batch, real-time, asynchronous, serverless), and the three learning paradigms (supervised, unsupervised, reinforcement learning).
- Pay special attention to the difference between precision and recall, and practice identifying which one matters more in a given business scenario based on the relative cost of false positives versus false negatives.
- Review the table of AWS managed AI services (Transcribe, Translate, Comprehend, Lex, Polly, Rekognition, Textract) in Task 8 (AWS Services Quick Reference Guide), focusing on the "commonly confused with" column for each service.

- Re-read the AI/ML pipeline stages and MLOps concepts (monitoring, retraining, technical debt) in Task 3, and self-test using the ordering-style practice questions in Task 12.
- Review the criteria for choosing traditional ML versus foundation models, with particular attention to explainability and regulatory scenarios.

Recommended re-study time: 2-3 hours, followed by retaking the Domain 1 questions from Task 12 (Domain Practice Question Bank) before reattempting this full practice exam.

Domain 2: Fundamentals of GenAI

If you scored below 70% on Domain 2 questions, here is exactly what to review:

- Return to Task 4 (Domain 2 Study Module) and re-read the foundational vocabulary section: tokens, chunking, embeddings, vectors, context window, and context engineering. These terms appear throughout Domains 2 and 3, so gaps here will affect performance elsewhere.
- Review the FM lifecycle stages (data selection, pre-training, fine-tuning, evaluation, deployment, feedback) and practice reconstructing the correct order from memory.
- Re-study agentic AI concepts: the Model Context Protocol (MCP), memory management, tool usage, and workflow orchestration, along with the distinction between Strands Agents (the open-source framework) and Amazon Bedrock AgentCore (the managed production service).
- Review the advantages and disadvantages of GenAI (hallucination, nondeterminism, limited interpretability) and the documented factors used to select a GenAI model for a business use case.
- Revisit the AWS GenAI services landscape in Task 8: Amazon Bedrock, SageMaker AI, SageMaker JumpStart, Amazon Quick, Kiro, and the cost tradeoffs between on-demand token pricing and provisioned throughput.

Recommended re-study time: 2-3 hours, followed by retaking the Domain 2 questions from Task 12 before reattempting this full practice exam.

Domain 3: Applications of Foundation Models

If you scored below 70% on Domain 3 questions, here is exactly what to review:

- Domain 3 carries the most exam weight (28%), so a weak score here deserves priority attention. Return to Task 5 (Domain 3 Study Module) in full.
- Re-read the FM selection criteria (cost, modality, latency, multi-lingual support) and the effect of inference parameters, especially temperature, on model output.
- Review RAG in depth: what it is, why organizations use it, how Amazon Bedrock Knowledge Bases supports it, and which AWS services can store vector embeddings (Amazon OpenSearch Service, Amazon Aurora, Amazon Neptune, and Amazon RDS for PostgreSQL with pgvector).
- Re-study the FM customization spectrum from lowest to highest cost: in-context learning, RAG, fine-tuning, model distillation, and full pre-training, along with when each is appropriate.
- Review prompt engineering thoroughly: the three prompt components (context, instruction, negative prompt), the five core techniques (zero-shot, single-shot, few-shot, chain-of-thought,

prompt templates), and the four prompt-related risks (exposure, poisoning, hijacking, jailbreaking).

- Review fine-tuning data preparation practices (curation, governance, sizing, labeling, representativeness, RLHF) and the four automated evaluation metrics (ROUGE, BLEU, BERTScore, LLM-as-a-judge), making sure you can match each metric to its appropriate use case (summarization, translation, semantic similarity, nuanced quality judgment).

Recommended re-study time: 3-4 hours, given the domain's weight, followed by retaking the Domain 3 questions from Task 12 before reattempting this full practice exam.

Domain 4: Guidelines for Responsible AI

If you scored below 70% on Domain 4 questions, here is exactly what to review:

- Return to Task 6 (Domain 4 Study Module) and re-read the six core features of responsible AI: bias, fairness, inclusivity, robustness, safety, and veracity. Practice distinguishing between them using realistic scenarios.
- Review the AWS tools for detecting and addressing bias and supporting explainability: Amazon SageMaker Clarify, Amazon SageMaker Model Cards, Amazon Augmented AI (A2I), and SageMaker Model Monitor, paying close attention to which tool solves which specific problem.
- Re-study the relationship between bias, variance, overfitting, and underfitting, and how each affects model fairness and performance.
- Review the documented legal risks of GenAI (intellectual property infringement, loss of customer trust, hallucination-driven misinformation, end-user risk) and the dataset characteristics that support responsible AI (inclusivity, diversity, curated sources, balanced representation).
- Revisit the principles of human-centered design for explainable AI, including user feedback mechanisms and AI decision transparency.

Recommended re-study time: 1.5-2 hours, followed by retaking the Domain 4 questions from Task 12 before reattempting this full practice exam.

Domain 5: Security, Compliance, and Governance for AI Solutions

If you scored below 70% on Domain 5 questions, here is exactly what to review:

- Return to Task 7 (Domain 5 Study Module) and re-read the AWS Shared Responsibility Model, with particular attention to which responsibilities belong to AWS versus the customer for managed AI services like Amazon Bedrock.
- Review the AWS services used to secure AI systems: IAM, AWS KMS, Amazon Macie, AWS PrivateLink, and Amazon Bedrock Guardrails, along with the "commonly confused with" distinctions in Task 8 (especially Macie versus Inspector).
- Re-study the AI-specific security and privacy considerations: prompt injection, data leakage, output filtering and validation, audit trails, and toxicity detection.
- Review hallucination detection and grounding techniques: RAG grounding, output validation, and confidence scoring, and be able to explain how each reduces the risk of inaccurate AI output.

- Revisit the governance and compliance services (AWS Config, Amazon Inspector, AWS Audit Manager, AWS Artifact, AWS CloudTrail, AWS Trusted Advisor) and be sure you can distinguish AWS Artifact (downloading AWS's own reports) from AWS Audit Manager (building your own compliance evidence).
- Review data governance concepts (data residency, data lifecycle, data retention) and the Generative AI Security Scoping Matrix as a governance framework.

Recommended re-study time: 1.5-2 hours, followed by retaking the Domain 5 questions from Task 12 before reattempting this full practice exam.

PART 7: "THINGS TO MEMORIZE" VS. "THINGS TO UNDERSTAND"

Use this two-column reference to focus your remaining study time efficiently. Items in the left column are best handled through repetition and flashcards (Task 11). Items in the right column require working through scenarios until the reasoning feels natural, not just memorized.

Things to Memorize	Things to Understand
The five domain weights (20%, 24%, 28%, 14%, 14%)	Why Domain 3 (Applications of Foundation Models) carries the most exam weight, and what that implies for study time allocation
Exam facts: 65 questions, 90 minutes, 50 scored / 15 unscored, 700/1,000 passing score	The compensatory scoring model, and why a weak section score doesn't automatically mean failure
Service names and one-line purposes (Bedrock, SageMaker AI, Transcribe, Translate, Comprehend, Lex, Polly, Rekognition, Textract, etc.)	How to choose between commonly confused services (Bedrock vs. SageMaker AI, Macie vs. Inspector, CloudTrail vs. CloudWatch, Artifact vs. Audit Manager) based on a scenario
The four prompt engineering techniques by name (zero-shot, single-shot, few-shot, chain-of-thought)	When to apply each prompt engineering technique based on the task and the amount of guidance the model needs
The four automated FM evaluation metrics by name (ROUGE, BLEU, BERTScore, LLM-as-a-judge)	Why ROUGE suits summarization, BLEU suits translation, and LLM-as-a-judge suits nuanced quality judgments at scale
The six core responsible AI features (bias, fairness, inclusivity, robustness, safety, veracity)	How bias and variance specifically affect demographic groups and contribute to overfitting or underfitting
The four prompt engineering risks (exposure, poisoning, hijacking, jailbreaking)	How RAG actually works end-to-end (chunking, embedding, retrieval, grounding) and why it reduces hallucination
Definitions of tokens, chunking, embeddings, vectors, and context window	The cost and complexity tradeoffs across the FM customization spectrum (in-context learning, RAG, fine-tuning, distillation, full pre-training)
The AWS Shared Responsibility Model's basic split ("of the cloud" vs. "in the cloud")	How the shared responsibility model applies specifically to a managed AI service like Bedrock, including edge cases like agent security (AgentCore Identity/Policy)

Things to Memorize	Things to Understand
Vector-capable AWS database services (OpenSearch Service, Aurora, Neptune, RDS for PostgreSQL with pgvector)	Why an organization would choose one vector storage option over another based on existing infrastructure and use case
The four types of inferencing (batch, real-time, asynchronous, serverless)	When each type of inferencing is the right choice given latency, volume, and cost constraints
The distinction between Strands Agents and Amazon Bedrock AgentCore	The broader concept of agentic AI: how memory management, tool usage, and workflow orchestration combine to let an agent complete multi-step tasks autonomously
Governance/compliance service names (AWS Config, Amazon Inspector, AWS Audit Manager, AWS Artifact, AWS CloudTrail, AWS Trusted Advisor)	How an organization would actually apply the Generative AI Security Scoping Matrix to classify a real use case by risk level
The names of the four exam question types (multiple choice, multiple response, ordering, matching)	How to reason through a multiple-response question by eliminating clearly wrong options first, since there is no partial credit
Practice exam passing threshold used in this program (47/65, 73%)	Why a small accuracy improvement in a model doesn't always translate into greater business value, given cost and outcome tradeoffs

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End of Task 9 - Full-Length Practice Exam, Instructor / Answer-Key Edition. This instructor version contains all answers and explanations and should not be distributed directly to learners. Task 10 will produce the clean, answer-free learner version of this same 65-question exam. Document based on AIF-C01 Exam Guide Version 1.1 (April 30, 2026).